Anonymity and Neutrality in Classification Aggregation

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Outline

The setting

Goal: classify a set of m objects X into a set of ρ predefined categories: P
Difference with ML: aggregation based on individual's classifications

Definition

A Classification Aggregation Function (CAF) is a function that takes classifications as an input and outputs one.

Classification $c \in \mathcal{C}, c : X \to P$, profile $\boldsymbol{c} \in \mathcal{C}^N$, CAF $\alpha : \mathcal{C}^N \to \mathcal{C}$.

Surjectivity

All classifications should be surjective

Low rationality requirement

Some examples: Job assignment, Group identification, Classifying celestial objects...

Independence

Classical axiom for aggregation

- Unary Independence: For $x \in X$, $\alpha(\boldsymbol{c})(x)$ depends only on individuals' classifications of x.
- Binary independence: deciding to put two items together depends only on classifications of these items.

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Discussed in Pref Aggregation and JA Dietrich [2015]
Not always meaningful for classifying (KNN approach)

Left wing or right wing ?

Fun website by Théo Delemazure

Figure: Screenshot from the website

Neutrality

- Anonymity: α is stable to swaps of voters.
- Object neutrality: α is stable to swaps of objects.
- Category neutrality: ...

Neutrality

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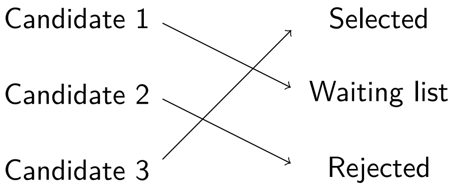
- Category neutrality: ...

Some motivations

- Anonymity and Object neutrality: ranked categories
- Anonymity and Category neutrality: Items allocation from voting
- Not always relevant: Fioravanti [2024]

Ranked categories

A jury needs to decide on selecting candidates,



Strong link with preference aggregation $\Rightarrow$ similar impossibility results.

Some results in Preference Aggregation

NB: These are simplified versions of the theorems

Theorem

Arrow [1951] Dictatorship is the only SWF that satisfies unanimity and binary independence.

Theorem

Wilson [1972] Dictatorship, anti dictatorship and null rule are the only SWF satisfying binary independence and citizen sovereignty.

Theorem

Moulin [1980, 1983], Bubboloni and Gori [2014] There can be an anonymous and object neutral SWF iff 1 is the only number smaller than m than divides n.

Arrovian impossibility

Theorem

Maniquet and Mongin [2016] Dictatorship is the only CAF that satisfies unary independence, unanimity and surjectivity.

	1	2	3
x	p	p	q
y	p	q	p
z	q	p	p

Table: Example where MAJ fails

A Wilsonian impossibility

Theorem

Cailloux et al. [2024] If a CAF satisfies unary independence, citizen sovereignety and surjectivity, it is essentially a dictatorship.

Example of essential dictatorship: take classification from 2 and apply $\pi = pq$

	1	2	3	result
x	p	p	q	q
y	p	q	p	p
z	q	p	p	q

Special case: 2 obects and categories

Drop surjectivity

Theorem

Craven [2023] If the domain of the aggregator is simple, MAJ satisfies unary unanimity, unary independence and binary unanimity.

Theorem

Craven [2023] If the domain of the aggregator is rich, then any aggregator that satisfies unary unanimity, unary independence, and binary unanimity is dictatorial.

Fuzzy Classifications

Theorem

Alcantud et al. [2019] Dictatorship is the only fuzzy CAF that satisfies unary independence, unanimity and fuzzy surjectivity.

Theorem

Fioravanti [2024] The arithmetic mean satisfies unary independence, unanimity and anonymity.

	Left W	Right W
wheelchair	60%	40%
chair	40%	60%
pizza	70%	30%

Table: A fuzzy classification

Fuzzy Classifications

Theorem

Alcantud et al. [2019] Dictatorship is the only fuzzy CAF that satisfies unary independence, unanimity and fuzzy surjectivity.

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Table: A fuzzy classification

A Moulinian impossibility result

Theorem

Working paper
There exists a surjective, anonymous and object neutral aggregator only if 1 is the only number smaller than m than divides n.

Example for $n = 4, m = 2$

	1	2	3	4
x	p	p	q	q
y	q	q	p	p

	1	2	3	4
x	q	q	p	p
y	p	p	q	q

Equal number of objects and categories

We use the relation with preference aggregation

Theorem

Working Paper
A CAF is anonymous and object neutral iff its isomorphic SWF is.

Corollary

When $m = \rho$, there can be an anonymous and object neutral aggregator iff 1 is the only number smaller than m than divides n.

About iterative plurality

Focus on anonymity and object neutrality $\rightarrow$ ordered categories

	1	2	3	4	5
x	p	p	q	q	p
y	q	q	p	p	p
z	q	p	p	p	q

Each object received as many votes as the others for each category $\rightarrow$ not resolute.

How to avoid ties

Special case: 5 voters, 3 objects and 3 categories
Only $3! = 6$ surjective classifications

	1	2	3	4	5
x	p	p	q	q	p
y	q	q	p	p	p
z	q	q	p	p	q

	1	2	3	4	5
x	p	p	q	q	p
y	q	q	p	p	p
z	p	q	p	q	q

Anonymity and Category neutrality

Use the relation with Preference Aggregation and Group Theory as in Bubboloni and Gori [2014]

Theorem (Working paper)

There can be an anonymous and category neutral CAF iff 1 is the only smaller number than ρ that divides n.

Anonymity and Category neutrality

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Theorem (Working paper)

There can be an anonymous and category neutral CAF iff 1 is the only smaller number than ρ that divides n.

Why is it only possible with category neutrality ?
When swapping categories, the classification always changes.

Experiments

Some open questions:

- Anonymity and object neutrality when $m > \rho$
- Anonymity, object and category neutrality

Greedy algorithm based on Nardi [2021], Doğan and Giritligil [2022]

Results for 2 categories

	2	3	4	5	6	7	8	9	10
2	no	yes	no	yes	no	yes	no	yes	no
3	no	no	no	yes*	no	yes*	no	no	no
4	no	no	no	yes*	no	yes *	no	no	no
5	no	no	no	no	no	?	no	no	no

Table: Possibility of having anonymous and neutral aggregators for $\rho = 2$ (n in columns, m in lines)

Results for 3 categories

	2	3	4	5	6	7	8	9	10
3	no	no	no	yes	no	yes	no	no	no
4	no	no	no	?	no	yes*	no	no	no
5	no	no	no	no	no	?	no	no	no

Table: Possibility of having anonymous and neutral aggregators for $\rho = 3$ (n in columns, m in lines)

What's next ?

- Anonymity and object neutrality
- Allowing for ties: multiple classifications or correspondences ?
- Dropping neutrality or anonymity
- Fair tie breaking mechanisms
- More expressive preferences

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